A Training-Free Dynamic Surface Distance Field for Efficient Finite-Element Contact of Deformable Bodies

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Abstract

Signed distance fields provide an attractive geometric representation for contact detection because distance, penetration depth, and contact normals can be obtained from local field queries. However, classical SDFs are typically static and therefore difficult to apply directly to deformable finite-element bodies whose contact surfaces evolve with the nodal degrees of freedom. Existing dynamic SDF approaches often rely on precomputed deformation snapshots, reduced bases, or neural approximations, which introduce training costs and generalization concerns.

This paper presents a training-free dynamic surface-distance framework for finite-element contact. Instead of reconstructing a deformed SDF from data, the proposed method induces a local oriented surface distance directly from the current finite-element boundary. The contact gap is defined as the signed distance from a slave surface sample to the current master surface, and the corresponding normal, closest-point projection, contact Jacobian, and penalty contact force are assembled consistently from the current FEM geometry. A standalone implementation is developed without dependence on Abaqus-generated matrices, force files, or post-processing outputs.

The implementation is validated on linear TET4 finite elements with normal penalty contact. Verification is organized in three layers: analytic and scikit-fem checks for the non-contact FEM stress and displacement fields, CalculiX contact-output comparisons for static C3D8 contact energy and dynamic C3D4 block-plane contact trajectories, and implementation-level closest-point checks for the accelerated distance query. In the final non-quick experiments, the dynamic FEM-induced surface-distance pipeline achieves speedups from $3.58\times$ to $57.07\times$ over brute-force all-triangle projection for the tested surface resolutions; a separate total-step timing check also remains faster than the brute-force counterpart in the tested cases. The current scope is limited to linear TET4 elasticity in the core solver, oriented local surface distances, spatial-hash broad phase, and frictionless penalty normal contact.

1 Introduction

Contact handling is a central component of deformable finite-element simulation. At each time step, a solver must identify potential contact pairs, evaluate separation or penetration, compute contact normals, and assemble force and stiffness contributions into the global system. These operations are geometric as well as mechanical: an inaccurate gap or normal can lead to incorrect contact activation, force direction errors, or unstable time integration. As surface resolution increases, the repeated closest-point search between deformable surfaces can become a significant computational bottleneck.

Signed distance fields offer an attractive representation for contact because a local query can provide distance, penetration depth, and normal direction. However, a conventional SDF is usually defined for a fixed shape. This assumption is not compatible with finite-element bodies whose boundary surfaces move with nodal degrees of freedom. A distance field constructed in the reference configuration generally does not remain a true Euclidean distance field after stretch, compression, or shear. Therefore, directly reusing a material-space SDF can introduce errors in both the gap value and the contact normal.

Dynamic or deformable SDF representations can address this issue by approximating how the distance field changes with deformation. Many such approaches rely on precomputed deformation examples, reduced-order bases, learned neural fields, or other data-driven approximations [2, 6]. These methods can be effective when the deformation space is well sampled, but they introduce offline preparation costs and may generalize poorly to deformation modes outside the training or basis space. For mechanics-oriented simulation, this creates a gap between the finite-element state, which is available at every step, and the distance representation, which may be only indirectly tied to the current surface geometry.

This paper proposes a training-free dynamic surface-distance framework for deformable FEM contact. Instead of reconstructing a deformed volumetric SDF from data, the method induces an oriented local distance directly from the current finite-element boundary surface. A spatial-hash broad phase first identifies candidate boundary triangles, and a local closest-point projection then evaluates the gap, closest point, barycentric weights, and surface normal on the current deformed surface. The resulting distance query is therefore tied directly to the current FEM configuration.

The current-surface distance is used consistently in the contact formulation. The contact gap is defined as the signed distance from a slave surface sample to the current master surface. The slave and master Jacobian blocks are assembled from the same normal and shape-function weights used in the closest-point projection. A frictionless penalty contact force and Gauss-Newton contact stiffness approximation are then assembled from this Jacobian. This gives a compact contact pipeline in which geometry evaluation and force assembly share the same local surface representation.

We implement the method in a standalone FEM-SDF prototype with internally assembled TET4 stiffness, mass, body-force, constraint, contact, and time-integration components. The core solver does not depend on Abaqus-generated matrices, force files, `.odb` files, or spreadsheet outputs. Experiments first verify the non-contact FEM backend against analytic stress/strain references and an equivalent scikit-fem model. The contact evidence then uses external CalculiX outputs: a static C3D8 contact-energy test validates force and energy replay on the deformed current surface, and a dynamic C3D4 block-plane case compares activation time, penetration, normal reaction, contact energy, center-of-mass motion, and contact-zone stress. Brute-force closest-point projection is retained only as an implementation-level geometry oracle and timing baseline. In the final non-quick experiments, the dynamic FEM-induced surface-distance pipeline achieves speedups from $3.58\times$ to $57.07\times$ over brute-force all-triangle projection for the tested surface resolutions.

In summary, this work makes four contributions. First, it introduces a training-free current-surface-induced distance representation for deformable FEM contact. Second, it derives a consistent contact gap, normal, Jacobian, and penalty force assembly from local current-surface projection. Third, it provides a standalone linear TET4 FEM-SDF implementation independent of Abaqus-generated solver artifacts. Fourth, it presents a reproducible validation and evidence package in which FEM field correctness is checked against analytic and scikit-fem references, contact correctness is checked against CalculiX contact outputs, and computational benefit is measured against brute-force all-triangle projection.

2 Related Work

Finite-element contact methods typically rely on geometric proximity queries between deformable surfaces, followed by the assembly of contact constraints or penalty forces [3]. Classical node-to-surface and mortar formulations compute gaps and normals from closest-point projections on the current or predicted contact surface [5]. These approaches are mechanically direct, but repeated all-pair or all-triangle closest-point projection becomes expensive as the number of candidate surface features grows. Our method keeps the current-surface projection principle, but combines it with a dynamic surface-distance query and spatial hashing to avoid brute-force all-triangle projection in the tested pipeline.

Signed distance fields have been widely used for collision detection and contact response because they provide a convenient way to query distance and normal information [7]. For rigid or fixed geometries, an SDF can be precomputed and queried efficiently during simulation. The difficulty arises when the body is deformable: the contact surface changes with the finite-element state, while a static reference SDF does not generally preserve current-configuration Euclidean distances. Our material-space baseline experiments show this issue directly under stretch and shear deformation, where the carried reference-space distance produces nonzero gap and normal errors.

Several dynamic SDF approaches address deformation by learning or approximating the evolution of the distance field from examples, reduced coordinates, or neural representations [2, 6]. These methods can provide fast queries after training, but their accuracy depends on the deformation space represented by the data or model. In contrast, the method proposed here does not train a distance model and does not require deformation snapshots. The distance is induced at query time from the current FEM boundary, which makes the geometric representation directly consistent with the nodal configuration used by the mechanics solver.

Efficient contact pipelines usually separate broad-phase candidate detection from narrow-phase geometric evaluation. Common broad-phase structures include uniform grids, spatial hashes, bounding-volume hierarchies, and related acceleration structures [1]. The current implementation uses a uniform spatial hash over padded triangle AABBs. This choice is simple and deterministic, and it is sufficient to demonstrate the paper's core comparison against brute-force all-triangle projection. We do not claim superiority over optimized production BVH, IPC collision pipelines, GPU collision detection, or commercial contact implementations [4].

The proposed method sits between traditional closest-point contact and precomputed or learned SDF contact. Like closest-point FEM contact, it evaluates gap and normal from the current deformed surface. Like SDF-based contact, it presents the narrow-phase query as a local distance evaluation. Unlike static or data-driven SDF approaches, it does not rely on a reference field, deformation training data, neural approximation, or external Abaqus-generated matrices. The result is a training-free and reproducible FEM-SDF contact prototype for the specific setting of linear TET4 elasticity and frictionless penalty normal contact.

3 Method

3.1 Current finite-element surface

Let the deformable body occupy a reference domain Ω_0, discretized by TET4 elements. The current nodal coordinates are denoted by $\mathbf{x}_a$. A boundary triangle e on the current surface is represented as

$$\mathbf{s}_e(\boldsymbol{\xi}, \mathbf{q}) = \sum_{a \in I_e} N_a^\Gamma(\boldsymbol{\xi}) \mathbf{x}_a, \tag{1}$$

where $\boldsymbol{\xi}$ denotes surface coordinates, N_a^Γ are the surface shape functions, and $\mathbf{q}$ collects the nodal degrees of freedom. The outward unit normal on the current surface patch is

$$\mathbf{n}_e = \frac{\mathbf{s}_{,\xi} \times \mathbf{s}_{,\eta}}{\|\mathbf{s}_{,\xi} \times \mathbf{s}_{,\eta}\|}. \tag{2}$$

The implementation repairs negatively oriented TET4 elements and orients boundary faces outward for closed TET4 surfaces, ensuring a consistent local sign convention for the supported geometries.

3.2 Dynamic surface distance

For a slave query point $\mathbf{x}_i^A$, the master body B defines a current surface-induced distance

$$g_i = \phi_B(\mathbf{x}_i^A, \mathbf{q}_B). \tag{3}$$

In the implementation, ϕ_B is evaluated by first obtaining candidate boundary triangles from a spatial hash broad phase and then computing the closest point on the candidate current-surface triangles. If $\mathbf{p}_i^*$ is the closest point and $\mathbf{n}_i$ is the corresponding oriented surface normal, the local signed gap is evaluated as

$$g_i = (\mathbf{x}_i^A - \mathbf{p}_i^*) \cdot \mathbf{n}_i. \tag{4}$$

A positive gap denotes separation, while a negative gap denotes penetration. This formulation should be interpreted as an oriented local surface distance induced by the current FEM surface. It is not claimed to be a robust global signed distance method for arbitrary open, non-manifold, or inconsistently oriented geometries.

3.3 Contact Jacobian

Let the slave query point be interpolated from slave body nodes as

$$\mathbf{x}_i^A = \sum_{b \in I_A} N_b^A \mathbf{x}_b^A. \tag{5}$$

Let the closest master point be

$$\mathbf{p}_i^* = \sum_{a \in I_B} N_a^B \mathbf{x}_a^B. \tag{6}$$

The first variation of the gap is

$$\delta g_i = \mathbf{n}_i^T \delta \mathbf{x}_i^A - \mathbf{n}_i^T \delta \mathbf{p}_i^*. \tag{7}$$

Therefore,

$$\frac{\partial g_i}{\partial \mathbf{x}_b^A} = N_b^A \mathbf{n}_i^T, \tag{8}$$

and

$$\frac{\partial g_i}{\partial \mathbf{x}_a^B} = -N_a^B \mathbf{n}_i^T. \tag{9}$$

These expressions define the contact Jacobian row $\mathbf{J}_i$.

3.4 Penalty contact force

For frictionless normal contact, a penalty energy is used:

$$E_{c,i} = \frac{1}{2} k_i \langle -g_i \rangle_+^2, \tag{10}$$

where

$$\langle x \rangle_+ = \max(x, 0). \tag{11}$$

The normal contact multiplier is

$$\lambda_i = k_i \langle -g_i \rangle_+. \tag{12}$$

The global contact force contribution is then

$$\mathbf{f}_{c,i} = \mathbf{J}_i^T \lambda_i. \tag{13}$$

A Gauss-Newton approximation to the contact stiffness may be assembled as

$$\mathbf{K}_{c,i} \approx k_i \mathbf{J}_i^T \mathbf{J}_i. \tag{14}$$

4 Implementation

The method is implemented as a standalone Python prototype. The solver contains modules for mesh topology, TET4 finite-element assembly, boundary extraction, current-surface distance queries, contact broad phase, contact narrow phase, penalty force assembly, and validation scripts.

The finite-element implementation supports small-strain linear TET4 elasticity. The element stiffness is assembled as

$$\mathbf{K}_e = V_e \mathbf{B}_e^T \mathbf{C} \mathbf{B}_e, \tag{15}$$

where V_e is the TET4 volume, $\mathbf{B}_e$ is the strain-displacement matrix, and $\mathbf{C}$ is the isotropic linear-elastic constitutive matrix. Both consistent and lumped mass options are implemented, and gravity/body-force assembly is performed internally.

The broad phase uses a uniform spatial hash over padded triangle AABBs. Candidate triangles are passed to the dynamic surface-distance query. A brute-force all-triangle closest-point projection is retained only as an implementation-level geometry oracle and timing baseline; external contact-solver outputs provide the main contact-correctness evidence.

The implementation is independent of Abaqus in the core solver. No Abaqus `.inp` parser, Abaqus-generated global stiffness/mass matrices, `.mat` force files, `.odb` files, or spreadsheet outputs are required by the core `src/sfc` package. The final Phase-6 experiment package confirms that no `src/sfc` files were changed during final result packaging.

5 Experiments and Results

The experiments are organized to separate mechanics correctness, contact correctness, and acceleration evidence. This separation is important because the proposed contribution is not a replacement for FEM mechanics; it is a current-surface distance query used inside a contact formulation. Therefore, the non-contact displacement and stress fields are first checked against analytic and external FEM references. Contact correctness is then evaluated against external solver contact outputs, primarily CalculiX reaction, contact energy, penetration, active contact count, and stress diagnostics. Brute-force closest-point projection is used only as an implementation-level geometry oracle and as the performance baseline for contact search cost.

The final paper-scale performance and material-space baseline results are taken from the Phase-6 non-quick experiment package under `results/paper_final`. Physical-comparison evidence is generated by the Phase-7 package and the locked paper numerical experiments under `paper/numerical_experiments`. The recorded Phase-6 runtime metadata is Python 3.13.5, NumPy 2.1.3, SciPy 1.15.3, Matplotlib 3.10.0, Windows 11 AMD64, and 24 logical CPUs.

Table 1: Validation roadmap. External solver outputs provide the primary contact-correctness evidence; brute-force closest-point projection is restricted to implementation checks and timing baselines.

Claim family	Evidence source	Role in the paper
FEM stress/displacement	Analytic patch and scikit-fem	Non-contact mechanics correctness
Static contact law and energy	CalculiX `contactenergy.inp`	External contact-output reference
Dynamic contact trajectory	CalculiX generated C3D4 block-plane case	External trajectory reference
Current-surface SDF geometry	CalculiX-deformed geometry replay	Gap/force/energy on deformed surface
Query acceleration	Brute-force all-triangle projection	Timing baseline only

Table 2: Analytic stress/strain patch verification for affine small-strain linear elasticity.

Resolution	Elements	Max strain error	Max stress error
1	5	0.000000×10^0	0.000000×10^0
2	40	0.000000×10^0	0.000000×10^0
3	135	0.000000×10^0	0.000000×10^0
4	320	0.000000×10^0	0.000000×10^0

5.1 FEM field verification before contact

We first verify the FEM field quantities before evaluating any contact claim. For an affine uniaxial patch, the TET4 strain energy, engineering strain, and stress are compared against analytic small-strain linear elasticity. The maximum reported relative energy error in the Phase-6 package is

$$3.47 \times 10^{-16}. \tag{16}$$

For $E = 2.0 \times 10^5$, $\nu = 0.25$, and affine displacement $u_x = 10^{-3}x$, $u_y = u_z = 0$, the analytic Voigt strain is $[10^{-3}, 0, 0, 0, 0, 0]$, and the analytic stress is $[240, 80, 80, 0, 0, 0]$. Table 2 shows that the recovered strain and stress match the analytic values to reported precision.

As a second mechanics check, a fixed-base cantilever is solved over mesh refinement and against scikit-fem on the same TET4 mesh. The internal refinement trend in Table 3 is included to show expected displacement/stress stabilization, while the external scikit-fem comparison in Table 4 checks the assembled stiffness, displacement, strain, stress, and von Mises fields against an independent implementation. Stress-cloud figures are used only for visual audit; quantitative comparisons use the element stress values in the CSV data.

The maximum von Mises relative error in this comparison is 7.996486×10^{-13}. Figure 5 compares the SFC and scikit-fem stress clouds for the finest tested mesh and shows the absolute stress error. These results verify that the linear TET4 stiffness assembly and resulting stress field agree with an external open-source FEM implementation for the tested equivalent model.

5.2 External static contact reference: CalculiX C3D8 contact energy

The first contact-correctness experiment uses CalculiX's official `contactenergy.inp` surface-to-surface contact test. CalculiX performs the static C3D8 contact solve and prints contact force and

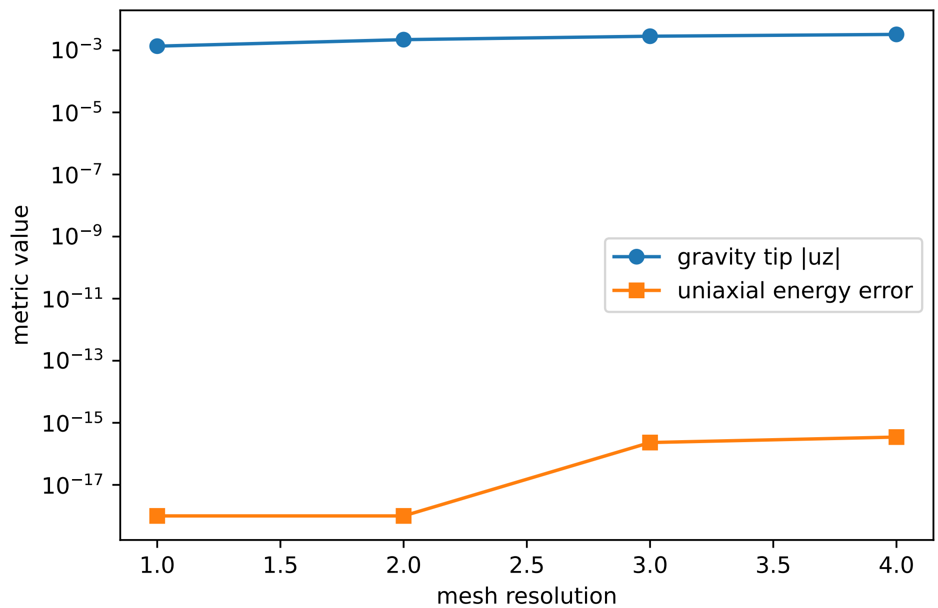

Figure 1: Mesh-resolution trend evidence generated by the Phase-6 package. The plot reports the linear-elastic patch energy error and the gravity-block displacement trend for the tested resolutions.

Table 3: Fixed-base cantilever displacement and von Mises stress trend. Relative errors are measured against the finest internal TET4 run.

Resolution	Elements	Tip u_z	Max von Mises	Tip rel. error
1	5	-1.397105×10^{-3}	9.796707×10^{1}	8.538791×10^{-1}
2	40	-3.686769×10^{-3}	1.835477×10^{2}	6.144070×10^{-1}
3	135	-6.617482×10^{-3}	2.557485×10^{2}	3.078887×10^{-1}
4	320	-9.561298×10^{-3}	3.144287×10^{2}	0.000000×10^{0}

contact spring energy quantities. SFC then replays the final CalculiX-deformed geometry with the proposed current-surface dynamic SDF on triangulated C3D8 boundary faces. This is an external contact-output comparison: the reference values are CalculiX force and contact-energy outputs, while brute-force projection is not used as the correctness target.

The replay reproduces the CalculiX normal force and contact spring energy to approximately 4×10^{-8} relative error. Because this case uses C3D8 elements, it also checks that the current-surface distance query is not topologically restricted to TET4 boundaries. The C3D8 mechanics solve used for visual comparison is validation-only; the main core solver remains the TET4 FEM-SDF implementation.

For a direct field comparison on the same C3D8 model, we additionally assemble a validation-only SFC C3D8 static solve with the same linear elastic material, constraints, nodal loads, and linear pressure-overclosure stiffness. This backend is used only for the external evidence case. It allows the SFC C3D8 von Mises field to be plotted against the CalculiX field and exposes the remaining error from the simplified validation contact discretization.

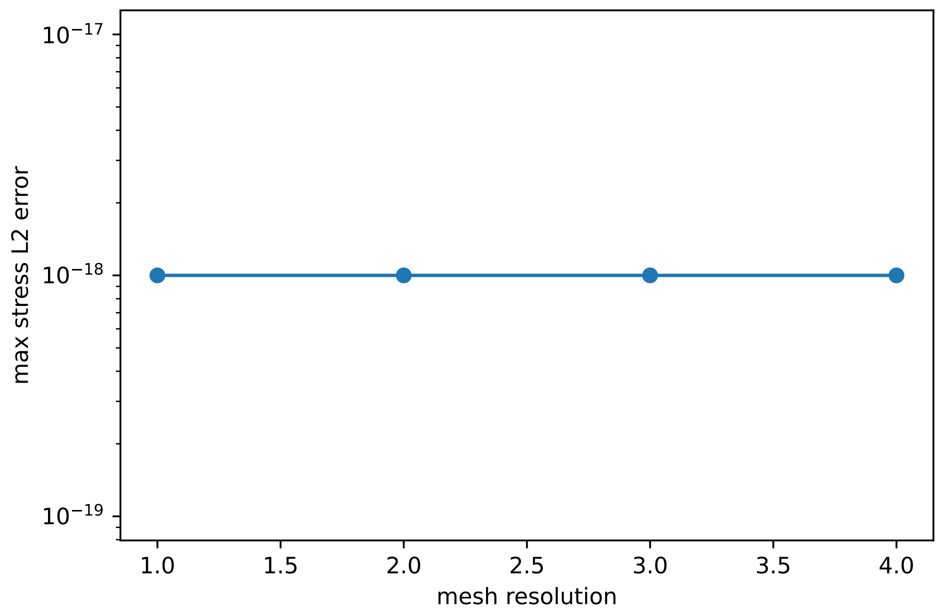

Figure 2: Stress recovery error for the analytic affine strain patch. The error floor reflects exact recovery of the affine field in the tested TET4 meshes.

Table 4: External open-source FEM comparison against scikit-fem 12.0.1 for the equivalent linear TET4 cantilever model.

Resolution	Elements	Disp. rel. error	Stress rel. error	K rel. error
1	5	1.733362×10^{-14}	1.730092×10^{-14}	3.626491×10^{-16}
2	40	5.060386×10^{-14}	5.730537×10^{-14}	5.197130×10^{-16}
3	135	1.095487×10^{-13}	1.276845×10^{-13}	2.713819×10^{-16}
4	320	8.203661×10^{-13}	8.507136×10^{-13}	6.519882×10^{-16}

5.3 External dynamic contact trajectory: CalculiX C3D4 block-plane

The second contact-correctness experiment compares a dynamic block-plane contact trajectory against CalculiX. The model uses a generated C3D4 block with matching material, time step $dt = 0.001$, duration 1 s, direct dynamics, a hard linear pressure-overclosure law, and the SFC `persistent_dynamic_sdf_calculix_f2f` contact mode. The comparison is intentionally scoped: it does not claim source-level equivalence to every private CalculiX branch, but it does compare the main physical trajectory and contact outputs that can be exported reliably.

This dynamic comparison supports the scoped claim that, with the same external contact law and C3D4 face-to-face contact structure, the dynamic-SDF backend preserves the CalculiX contact trajectory to small errors in the tested block-plane case. The active contact lifecycle is also checked: CalculiX and SFC both reach a maximum CNUM-equivalent value of 14, and the replay-on-CalculiX-displacement CNUM sequence matches the CalculiX sequence in the locked diagnostic output.

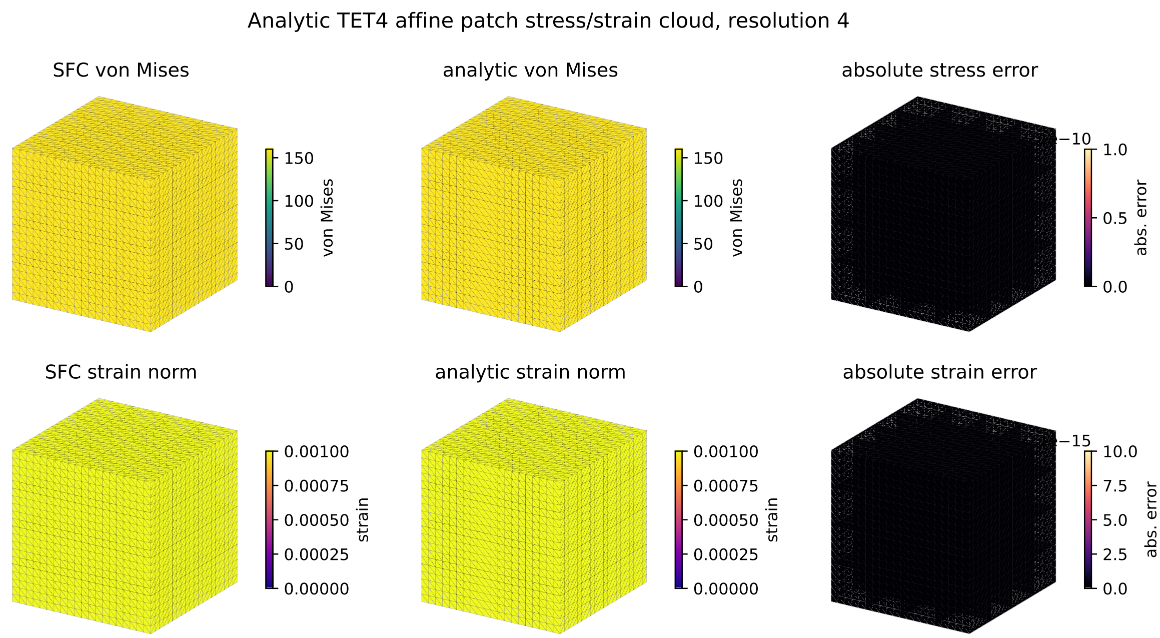

Figure 3: Analytic affine TET4 patch stress/strain cloud on the finest tested resolution. The first row compares SFC and analytic von Mises stress and shows the absolute stress error; the second row compares engineering strain norm and its absolute error. The uniform fields and near-zero error panels are expected for the affine patch.

Table 5: Static external contact-output comparison on CalculiX `contactenergy.inp`. SFC replays the final CalculiX-deformed C3D8 boundary with the current-surface dynamic SDF.

Quantity	CalculiX	SFC dynamic-SDF replay	Relative error
Normal force magnitude	4.000000	3.999999848	3.797933×10^{-8}
Contact spring energy	2.000010×10^{-5}	$2.000009924 \times 10^{-5}$	3.798346×10^{-8}
Maximum penetration	–	1.000005×10^{-5}	–
Active quadrature rows	–	2	–

5.4 Implementation-level geometry oracle

The external contact experiments above are the main correctness evidence for contact behavior. Separately, we retain a brute-force all-triangle closest-point projection check as an implementation oracle for the SDF query itself. In the final packaged run, the reference used surface resolution 12, with 288 triangles and 144 reference rows. The maximum reported gap error and normal error were both zero for the tested configuration.

This confirms that the spatial-hash candidate pipeline and dynamic surface-distance query reproduce direct closest-point projection for the tested geometry. It does not by itself prove physical contact correctness, and it does not imply global SDF correctness for arbitrary non-manifold or inconsistently oriented geometry.

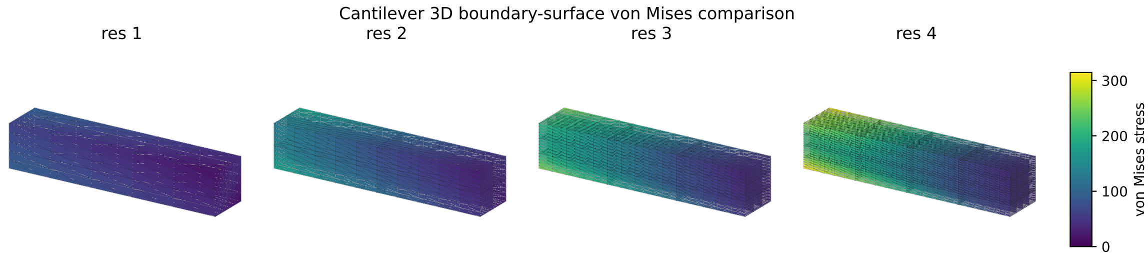

Figure 4: Standard FEM cantilever 3D boundary-surface stress comparison. Element von Mises stresses are averaged to nodes and interpolated over subdivided boundary triangles for visualization; the shared color scale is used across mesh resolutions.

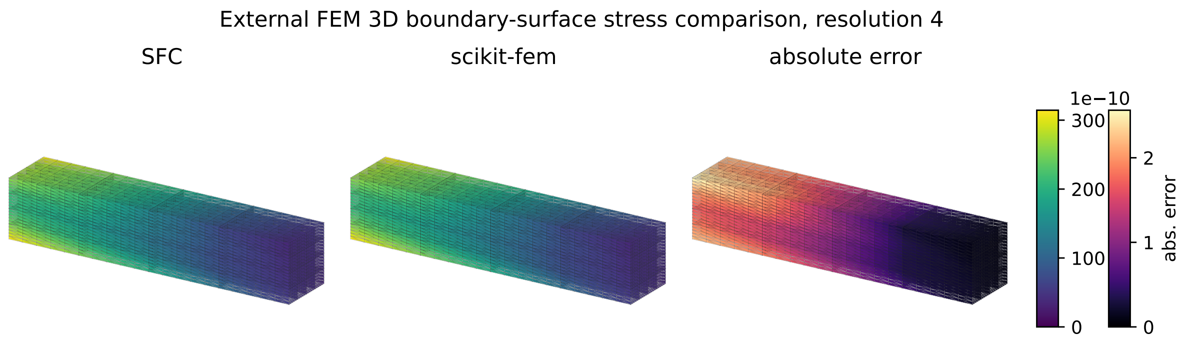

Figure 5: External FEM 3D boundary-surface stress comparison against scikit-fem on the finest tested equivalent cantilever model. The first two panels show nodal-averaged interpolated SFC and scikit-fem von Mises stress on the boundary surface; the third panel shows the interpolated absolute error.

5.5 Internal contact force-displacement regression

We also retain an internal force-displacement regression against brute-force all-triangle projection on a simple rigid-plane penalty case. Both paths use the same penalty law, so the comparison isolates whether the accelerated query produces the same active set and force for this implementation test. This table is not used as the main contact validation claim; the external CalculiX comparisons provide that role.

5.6 Material-space SDF baseline

A frozen material-space SDF baseline was evaluated under stretch and shear deformation. This baseline is not the proposed method; it is used only to demonstrate the errors introduced by carrying a reference-space distance function under deformation.

For stretch deformation, the maximum gap error was 5.217391×10^{-3}. For shear deformation, the maximum gap error was 2.708013×10^{-3}, and the maximum normal angle error was 3.580203×10^{-1}. These results support the motivation that a reference/material-space SDF is not sufficient to represent current-configuration distance and normal information under deformation.

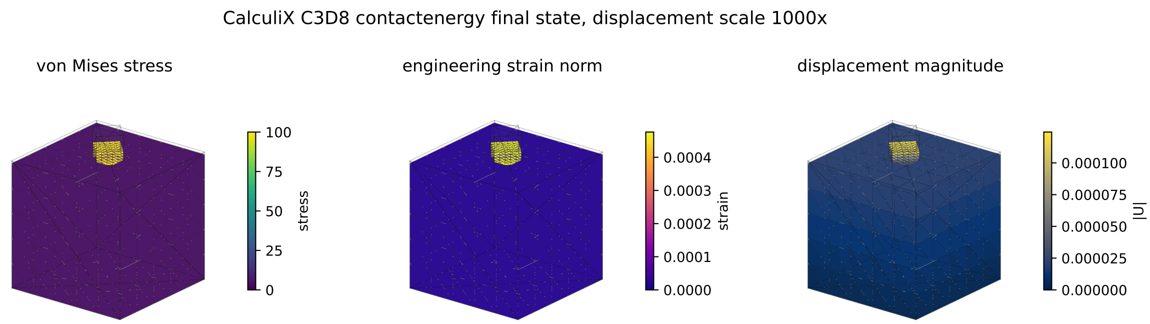

Figure 6: CalculiX `contactenergy.inp` C3D8 final-state visualization. The stress and strain fields are post-processed from the final CalculiX displacement field, averaged to boundary nodes, and interpolated over triangulated C3D8 boundary faces. The deformed shape is plotted with a labeled displacement scale factor to make the small static deformation visible.

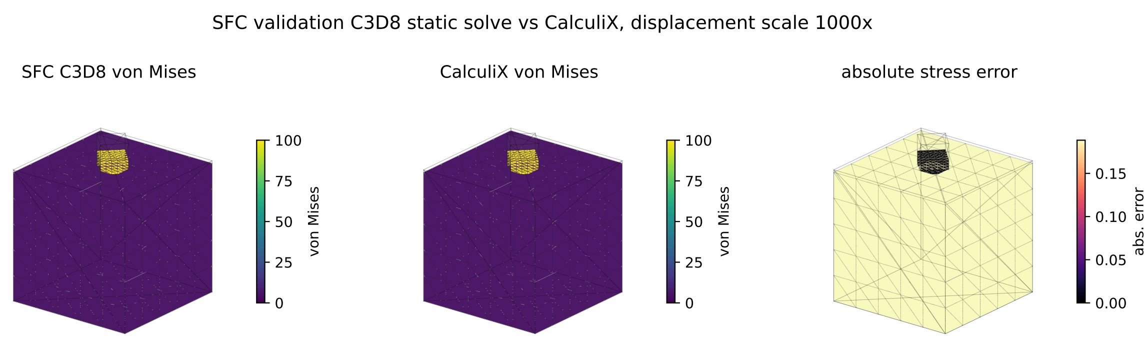

Figure 7: Direct validation-only C3D8 comparison on CalculiX `contactenergy.inp`. The first panel shows the SFC C3D8 static solve, the second shows CalculiX, and the third shows the absolute von Mises stress error. This comparison is included to make the external contact evidence visually auditable; it is not a claim that C3D8 is the main solver element type.

5.7 Contact time-history diagnostics

The internal contact diagnostics record time, minimum gap, maximum penetration, active contact count, normal force, contact energy, and action-reaction imbalance. These diagnostics are useful for regression testing and for separating geometry errors from force-assembly errors, but they are not a substitute for the external CalculiX contact-output comparisons above.

5.8 Performance scaling

The final non-quick experiments were run with surface sizes 4, 8, 12, and 16, using five timing repeats per row. The dynamic FEM-induced SDF full pipeline achieved the speedups over brute-force all-triangle projection shown in Table 9.

These results support the claim that, for the tested configurations, the dynamic FEM-induced surface-distance pipeline reduces contact detection cost relative to brute-force all-triangle projection. To check whether this remains meaningful beyond contact-only timing, Phase-7 also measures a small total-step workload that includes FEM assembly, a deterministic sparse linear solve, and

Table 6: Dynamic external contact trajectory comparison against CalculiX for the locked 1 s block-plane case.

Metric	CalculiX	SFC dynamic-SDF	Relative/absolute error
Contact activation time	2.200000×10^{-2}	2.200000×10^{-2}	1.39×10^{-17} abs.
Maximum penetration	1.044401×10^{-2}	1.068598×10^{-2}	2.316794×10^{-2}
Peak normal force	1.071145×10^{2}	1.081660×10^{2}	9.816751×10^{-3}
Peak contact energy	4.391052×10^{-1}	4.479852×10^{-1}	2.022301×10^{-2}
Maximum CNUM equivalent	14	14	0.000000 abs.
z_{cm} trajectory	–	–	2.601493×10^{-4} L2 rel.
Contact-zone max von Mises	4.570907×10^{1}	4.588905×10^{1}	3.937603×10^{-3}

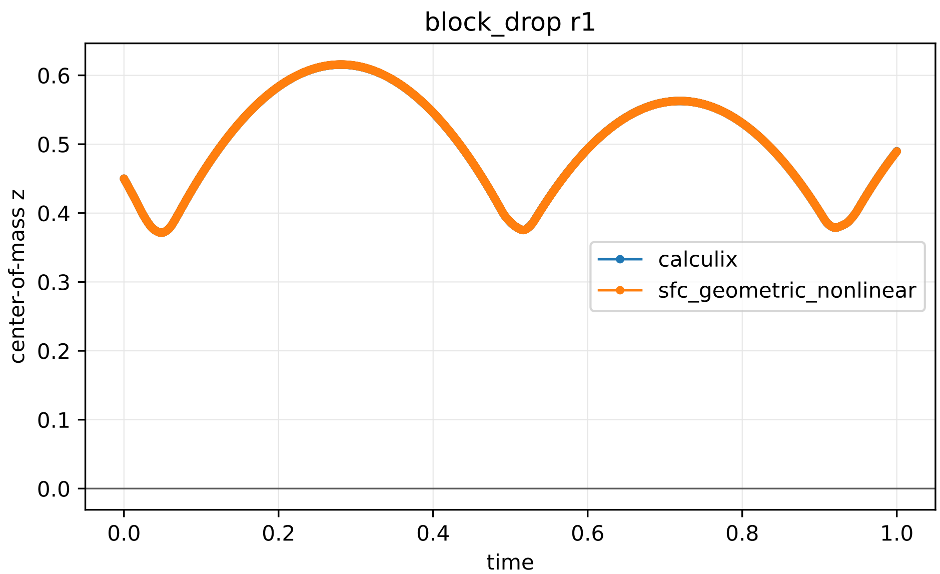

Figure 8: Dynamic block-plane center-of-mass trajectory compared against CalculiX for the locked 1 s contact case.

contact evaluation. Table 10 shows that the dynamic path remains faster than the brute-force counterpart for the tested sizes.

The comparison is not a claim of superiority over production BVH, IPC, GPU collision detection, or commercial solvers. It shows that, in the current prototype and tested workloads, replacing brute-force all-triangle projection with the spatial-hash dynamic SDF query can reduce both contact-only cost and the measured assembly/solve/contact step cost.

6 Limitations

The current implementation and experiments have several limitations.

First, the FEM formulation is limited to small-strain linear TET4 elasticity. Nonlinear materials, large-deformation FEM, corotational formulations, and hyperelasticity are not implemented.

Second, contact is limited to frictionless normal penalty contact. Friction, self-contact, augmented Lagrangian methods, active-set methods, barrier contact, and IPC-style formulations are

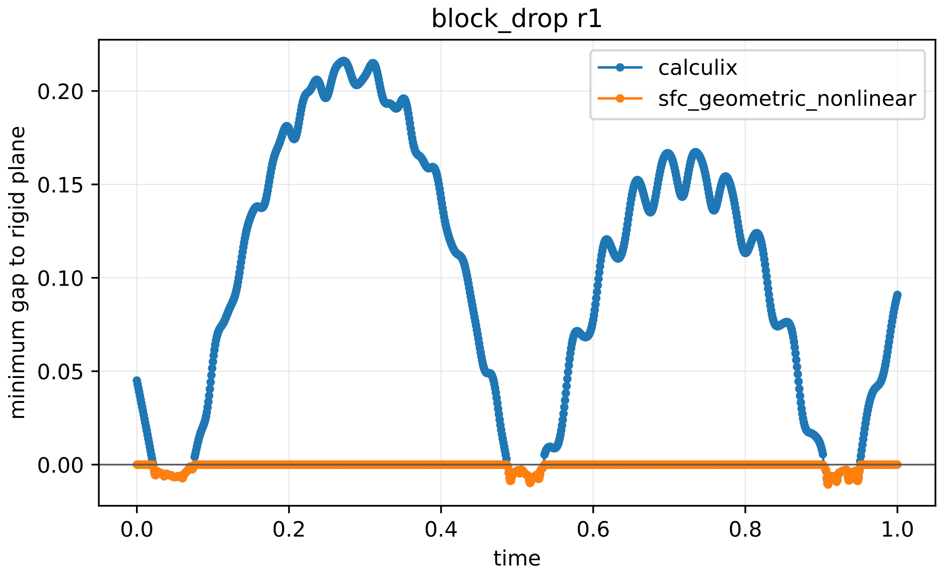

Figure 9: Minimum contact gap history in the dynamic CalculiX comparison. The plot reports the penetration behavior produced by the dynamic-SDF contact backend and the CalculiX reference output.

Table 7: Rigid-plane contact force-displacement comparison. The dynamic SDF path and brute-force CPP reference use the same normal penalty law.

Penetration	Active count	Brute-force force	Dynamic SDF force	Relative error
0.000000×10^{0}	0/0	0.000000×10^{0}	0.000000×10^{0}	0.000000×10^{0}
2.500000×10^{-3}	64/64	8.000000×10^{2}	8.000000×10^{2}	0.000000×10^{0}
5.000000×10^{-3}	64/64	1.600000×10^{3}	1.600000×10^{3}	0.000000×10^{0}
1.000000×10^{-2}	64/64	3.200000×10^{3}	3.200000×10^{3}	0.000000×10^{0}
2.000000×10^{-2}	64/64	6.400000×10^{3}	6.400000×10^{3}	0.000000×10^{0}

outside the current scope.

Third, the distance query is an oriented local current-surface distance. The method is validated for outward-oriented closed TET4 surfaces, but it is not a robust global signed-distance method for arbitrary open, non-manifold, or inconsistently oriented geometries.

Fourth, the broad phase is a uniform spatial hash. The reported speedups are measured against brute-force all-triangle projection, not against optimized production BVH implementations.

Finally, timing results are hardware- and implementation-dependent. The final experiment metadata must accompany any performance table derived from these results. The reported numbers should be interpreted as evidence for the current standalone prototype and not as a universal performance bound.

The external evidence is also bounded. The stress/strain check is an analytic affine patch test, and the external open-source FEM comparison uses scikit-fem on matching linear TET4 cantilever models. The contact evidence uses CalculiX output for a static C3D8 contact-energy replay and a scoped dynamic C3D4 block-plane trajectory. These cases support the current-surface contact query and force/energy replay in the tested settings, but they do not establish general nonlinear

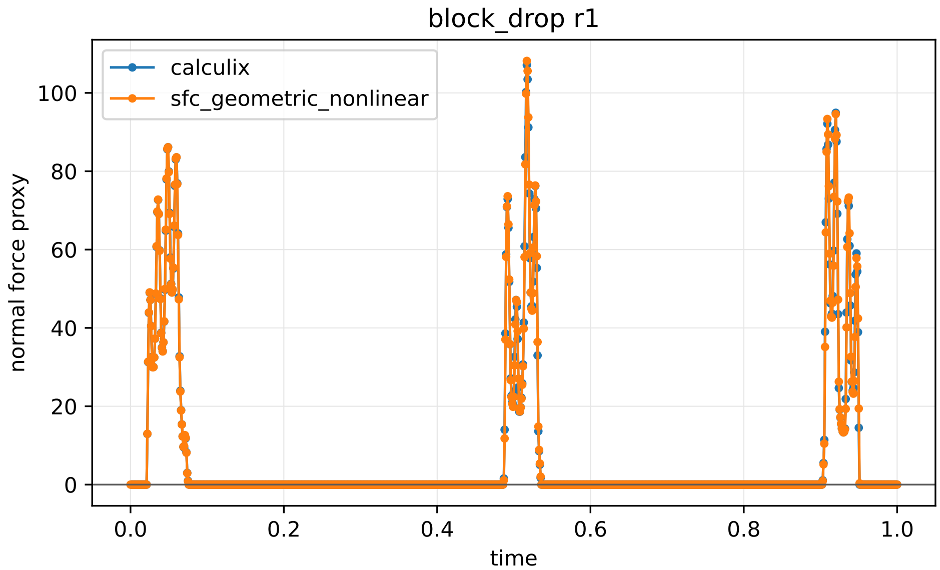

Figure 10: Normal reaction force history for the locked dynamic block-plane comparison. Force agreement is evaluated using CalculiX reaction/contact output where available.

Table 8: Material-space SDF baseline error under deformation. The baseline is a frozen material-space SDF and is used only for comparison.

Deformation	Rows	Max gap error	Max normal angle error
Stretch	20	5.217391×10^{-3}	0.000000×10^{0}
Shear	20	2.708013×10^{-3}	3.580203×10^{-1}

FEM, frictional contact, self-contact, sphere-drop impact, production BVH, or source-level CalculiX equivalence.

7 Conclusion

This paper presented a training-free dynamic surface-distance framework for finite-element contact of deformable bodies. Unlike static SDFs or data-driven dynamic SDF reconstruction, the proposed method induces a local oriented distance directly from the current finite-element boundary. The resulting contact gap, normal, Jacobian, and penalty force are assembled consistently from the current FEM geometry.

A standalone implementation was developed without dependence on Abaqus-generated matrices or data files. The prototype supports linear TET4 finite elements, spatial-hash contact candidates, dynamic surface-distance queries, local closest-point projection, and frictionless penalty contact. Validation includes analytic linear-elastic stress/strain references, external scikit-fem displacement/stress comparisons, CalculiX static contact-energy replay on a C3D8 surface, a CalculiX dynamic C3D4 block-plane trajectory comparison, material-space SDF baseline comparisons, implementation-level closest-point checks, and repeated performance timing.

In the tested final non-quick experiments, the dynamic FEM-induced surface-distance pipeline achieved 3.58× to 57.07× speedup over brute-force all-triangle projection. These results suggest

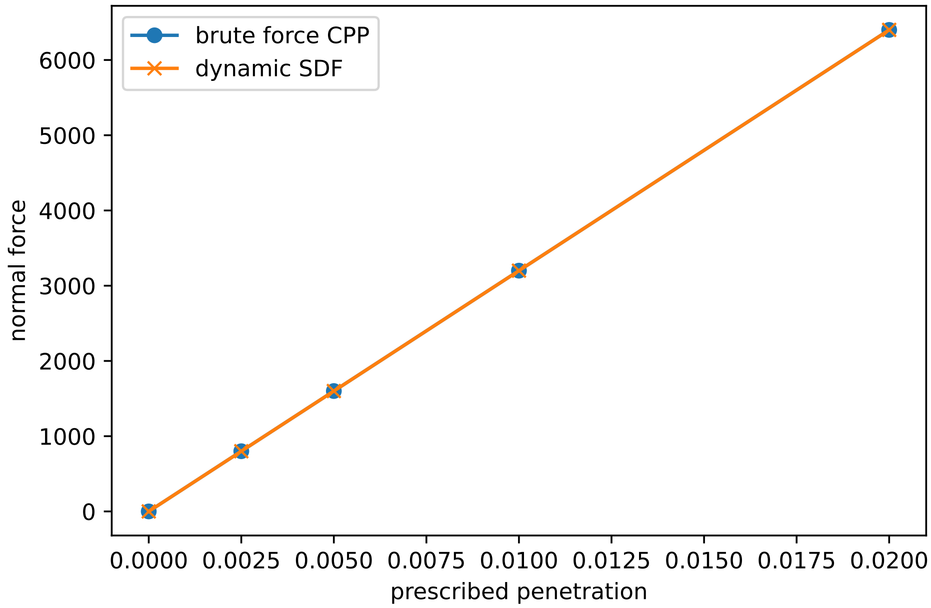

Figure 11: Contact force-displacement comparison against brute-force closest-point projection contact. The curves overlap for the tested rigid-plane normal penalty case.

Table 9: Final non-quick performance scaling. Speedup is measured against brute-force all-triangle projection, not against production BVH, IPC, GPU collision detection, or commercial solvers.

Surface res.	Triangles	Query points	Dynamic mean (s)	Speedup
4	32	16	1.13×10^{-2}	$3.58\times$
8	128	64	4.15×10^{-2}	$14.99\times$
12	288	144	9.40×10^{-2}	$32.48\times$
16	512	256	1.69×10^{-1}	$57.07\times$

that current-surface-induced dynamic distance queries can provide an efficient and training-free geometric front end for deformable FEM contact. Future work will extend the framework to nonlinear FEM, frictional and self-contact, robust global sign handling, production BVH/GPU acceleration, and barrier or augmented-Lagrangian contact formulations.

Evidence Package

The final experiment package is stored under `results/paper_final`. Figures are drawn from `results/paper_final/figures`, tables from `results/paper_final/tables`, and runtime metadata from `results/paper_final/metadata`. The added physical-comparison outputs are stored under `results/phase7`. The locked external evidence used in this draft is stored under `paper/numerical_experiments/scikit_fem_cantilever_external`, `paper/numerical_experiments/calculix_contactenergy_c3d8_replay`, and `paper/numerical_experiments/block_drop_dynamic_sdf_calculix_1s`. The source handoffs are documented in `docs/phase6_final_experiment_handoff.md`, `docs/phase7_physical_validation_handoff.md`, and the corresponding numerical-experiment manifests.

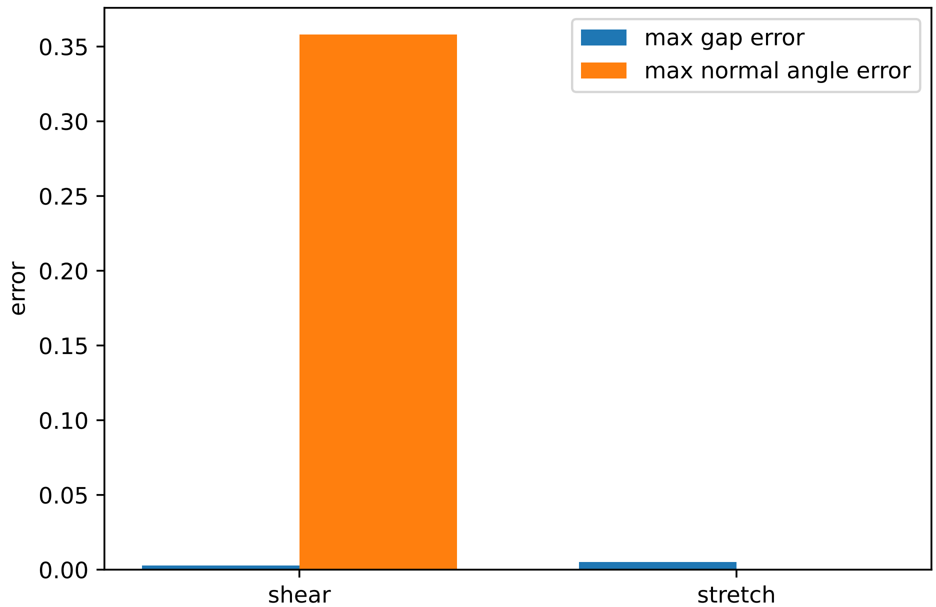

Figure 12: Material-space SDF baseline error under stretch and shear. The proposed method queries the current FEM surface; the baseline carries a frozen material-space distance.

Table 10: Phase-7 acceleration feasibility. Total-step timing includes FEM assembly, sparse linear solve, and contact evaluation.

Surface res.	Triangles	Contact-only speedup	Total-step speedup
4	32	4.50×	3.24×
8	128	15.55×	13.25×
12	288	33.72×	30.63×

Citation TODO

All placeholder bibliography entries in `references.bib` must be replaced before submission. Required citation groups include FEM node-to-surface contact, mortar contact, penalty contact, SDF contact, deformable or dynamic SDFs, neural/reduced SDFs, broad-phase collision detection, BVHs, and IPC/barrier contact.

References

[1] Todo: replace with broad-phase collision detection references. Placeholder citation for uniform grids, spatial hashing, and BVHs.

[2] Todo: replace with deformable or dynamic sdf references. Placeholder citation for dynamic/deformable signed distance fields.

[3] Todo: replace with finite-element contact references. Placeholder citation for FEM contact and node-to-surface contact.

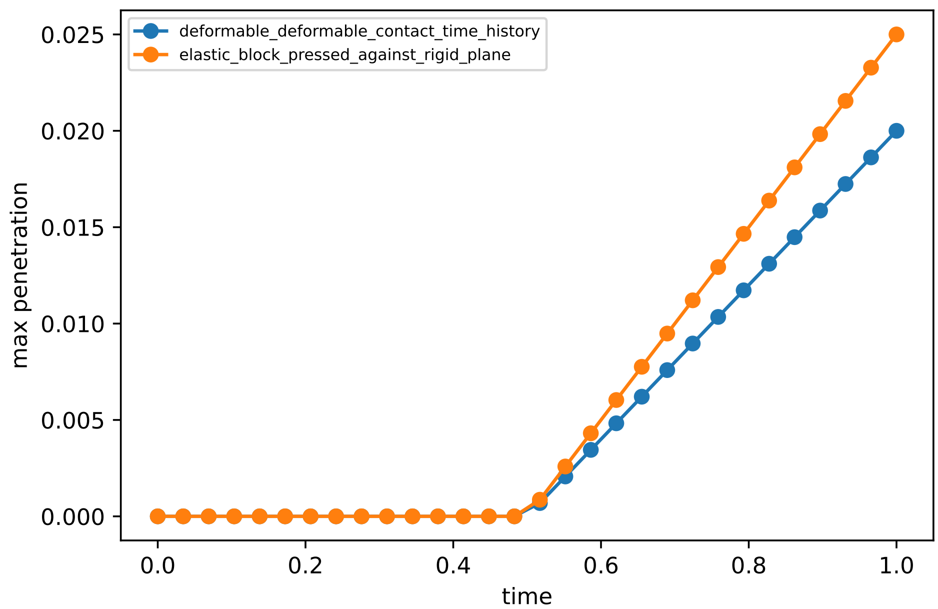

Figure 13: Contact time-history diagnostics from the Phase-6 package. The plotted quantities summarize penetration behavior for the rigid-plane and deformable-deformable contact checks.

[4] Todo: replace with ipc or barrier-contact references. Placeholder citation for IPC, barrier contact, and related out-of-scope methods.

[5] Todo: replace with mortar contact references. Placeholder citation for mortar contact and surface-to-surface contact.

[6] Todo: replace with neural or reduced sdf references. Placeholder citation for neural SDFs, reduced bases, and data-driven distance fields.

[7] Todo: replace with signed-distance-field contact references. Placeholder citation for signed distance fields in collision and contact.

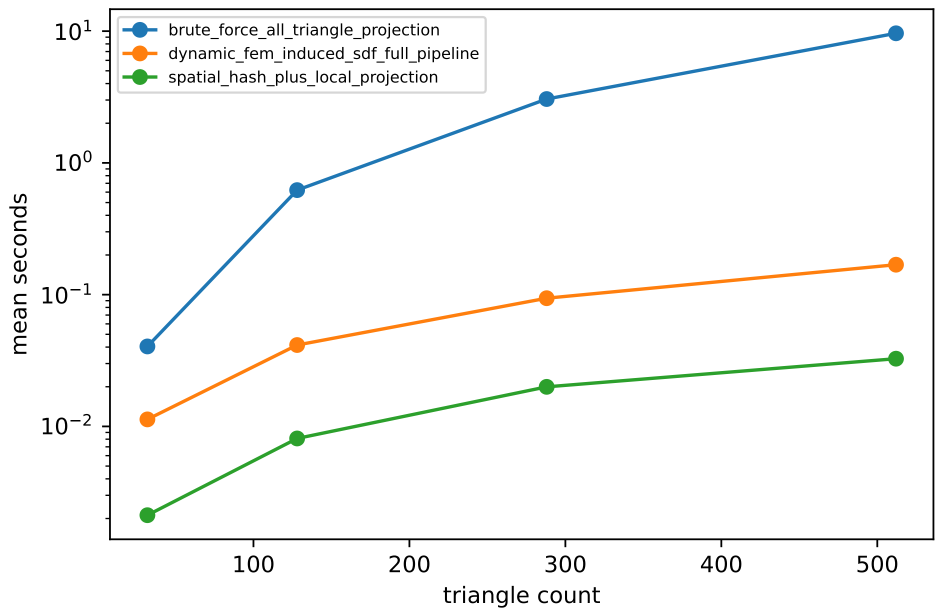

Figure 14: Performance scaling for brute-force all-triangle projection, spatial-hash plus local projection, and the dynamic FEM-induced SDF full pipeline.

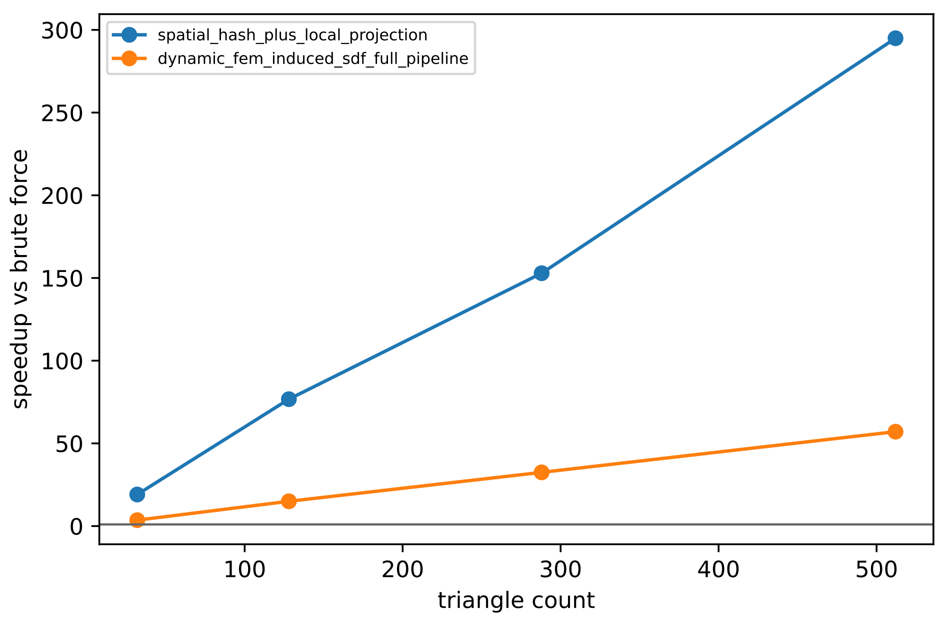

Figure 15: Measured speedup against brute-force all-triangle projection for the tested surface resolutions.

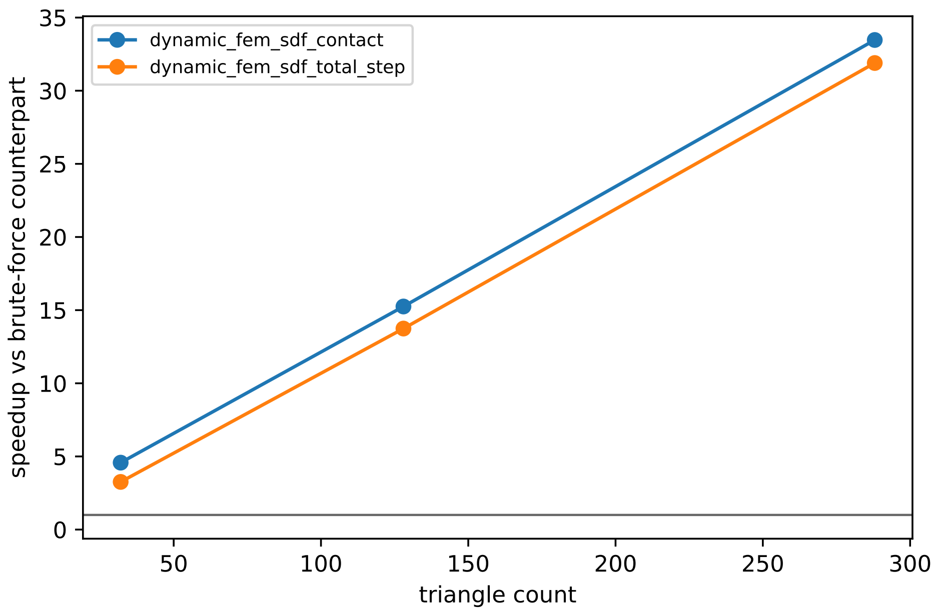

Figure 16: Phase-7 acceleration feasibility. The plot separates contact-only speedup from total-step speedup that includes FEM assembly and a sparse linear solve.